CS & IT ENGINEERING

DIGITAL LOGIC

MINIMIZATION



Lecture No. 4

By - CHANDAN SIR

Recap of Previous Lecture



K-MAP

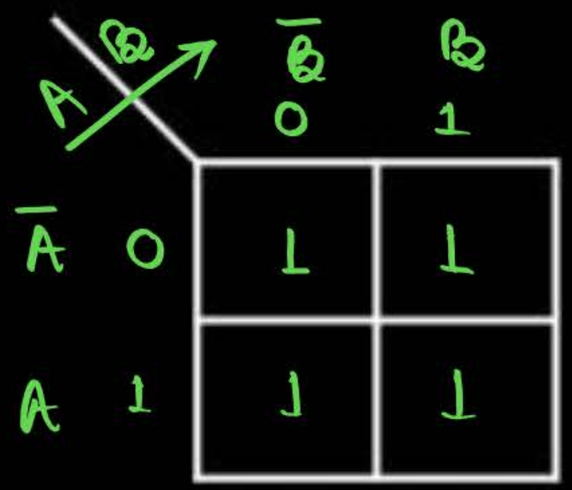
Topics to be Covered



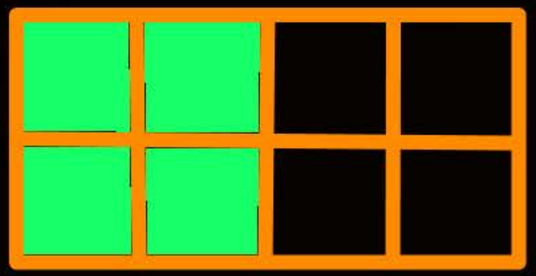
K-Map

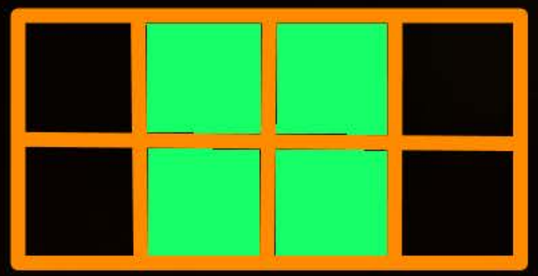
- Questions Practice
- Discussion

#Q. $f(A, B) = \overline{A}\overline{B} + \overline{A}B + A\overline{B} + AB = \Sigma m(0,1,2,3)$

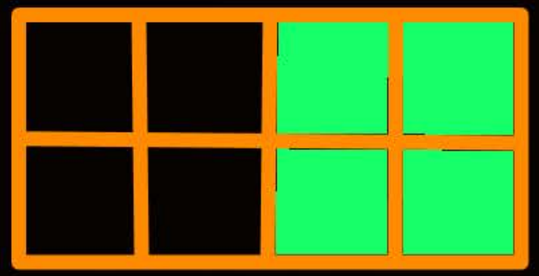


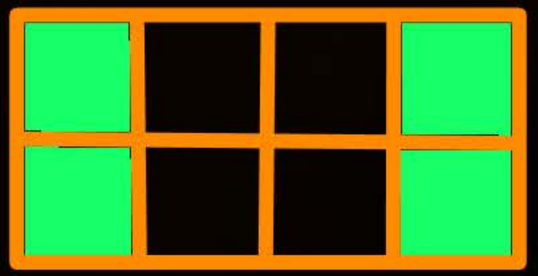
$= 1$

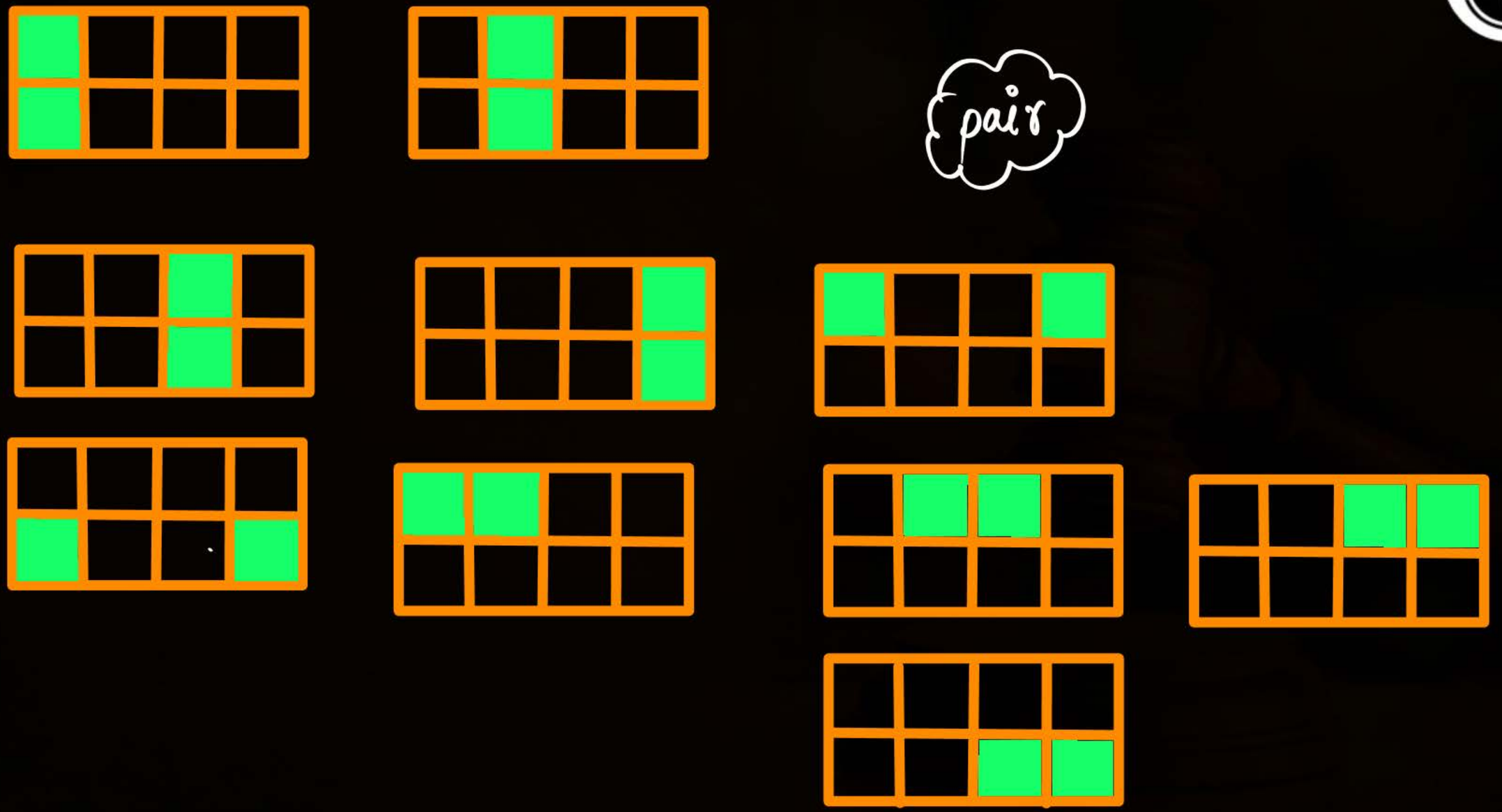




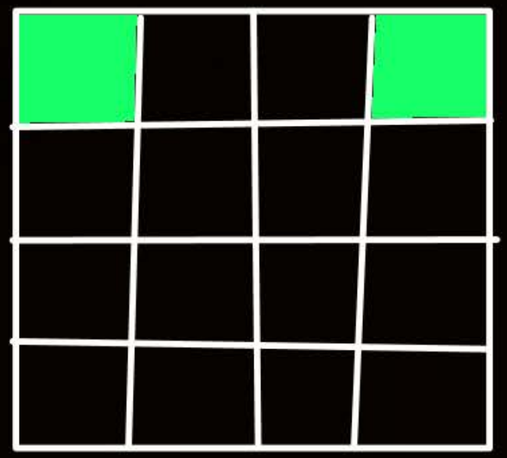
Quad

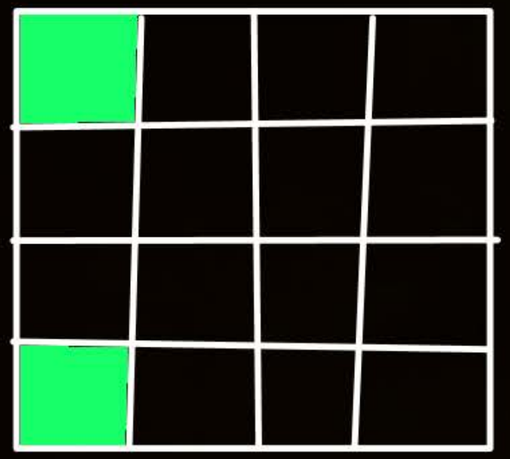


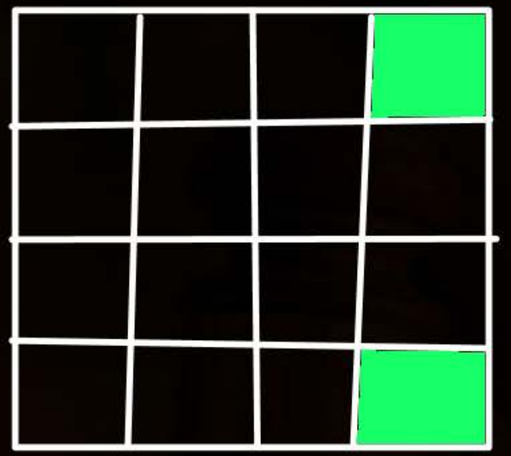


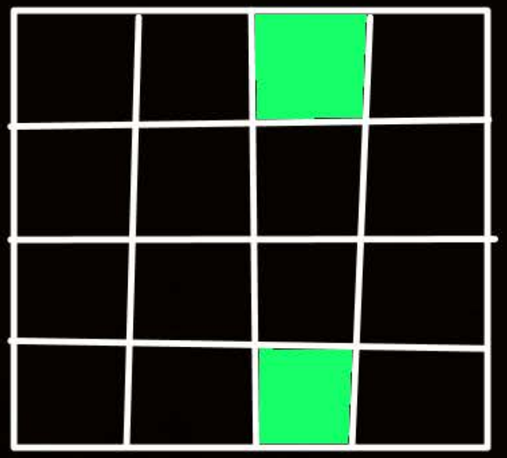

pair

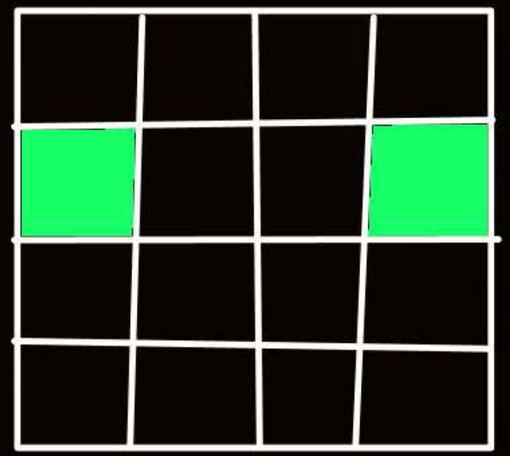
Pair

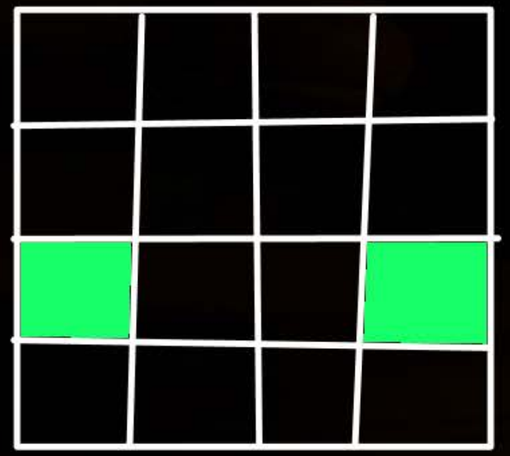




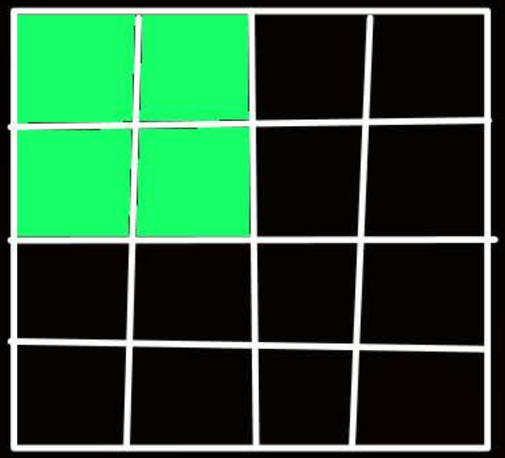


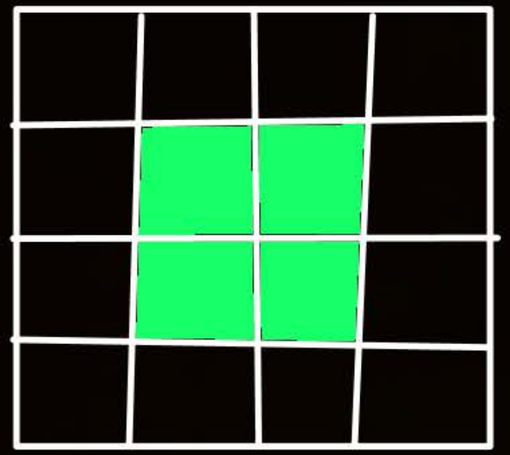


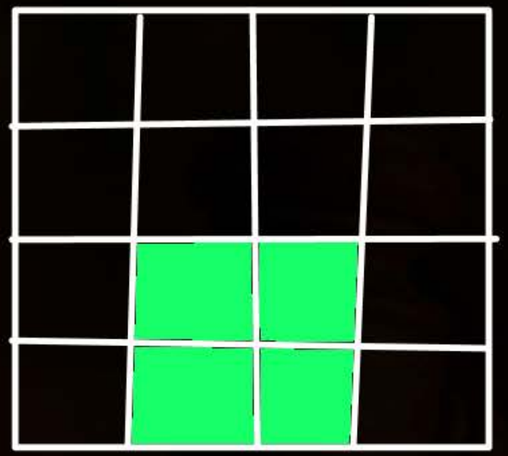


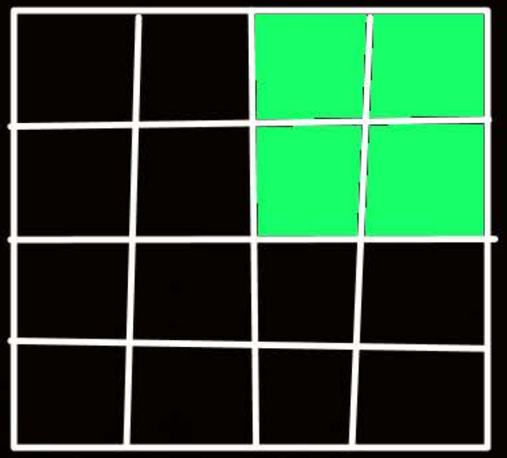


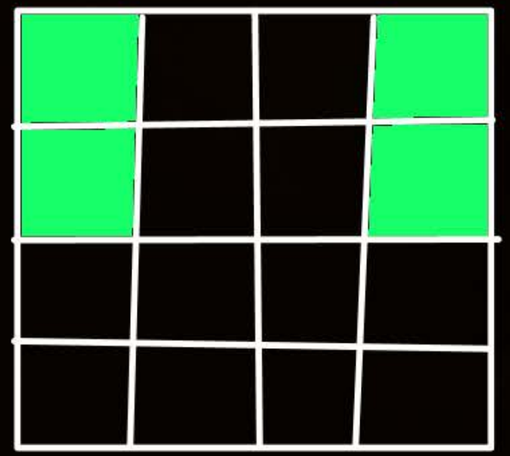
Quad

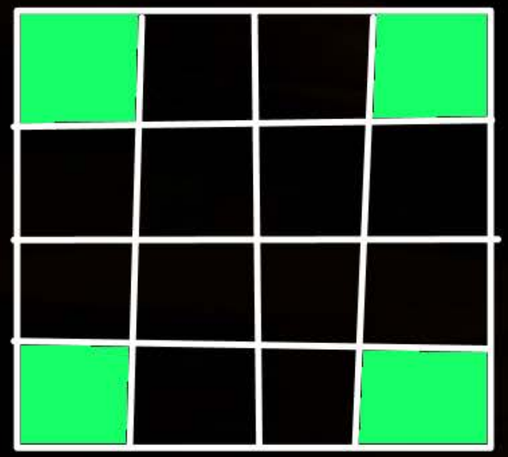


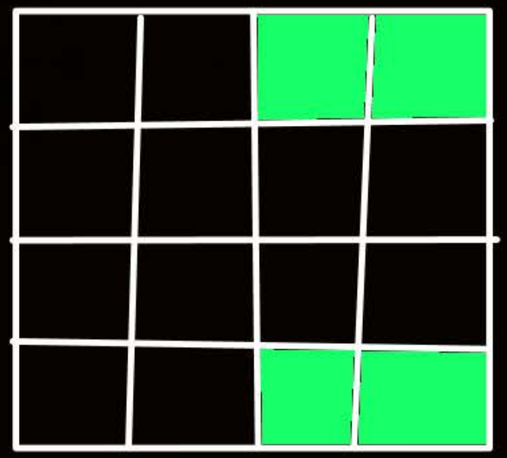
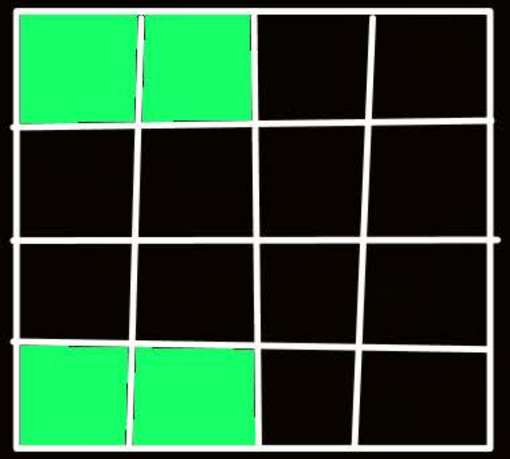
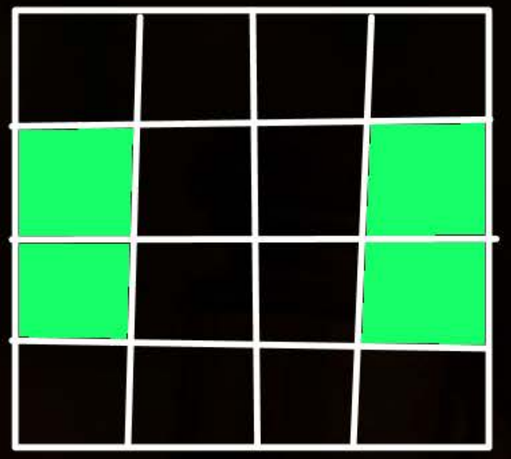
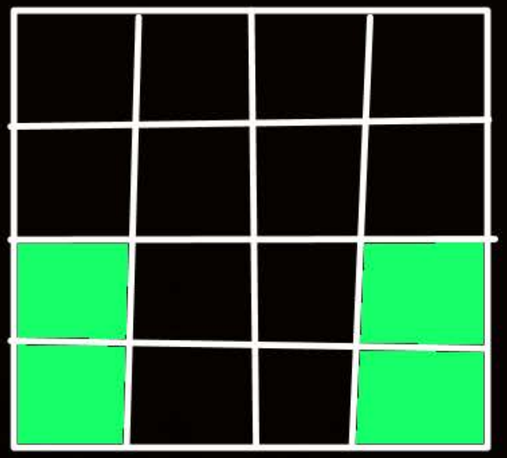
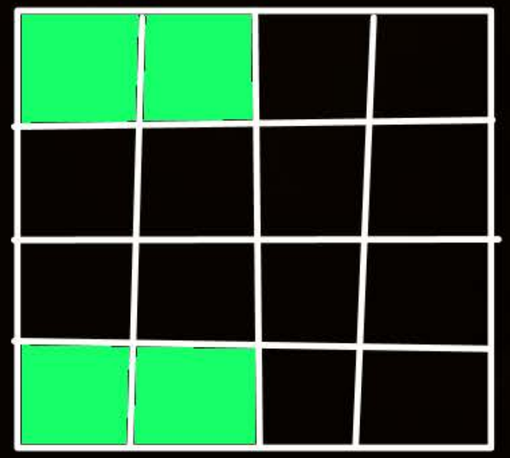
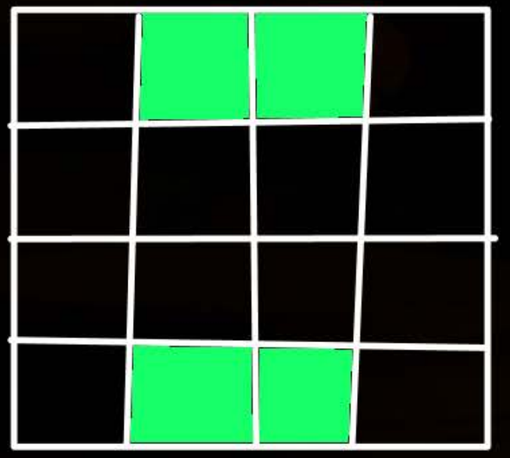


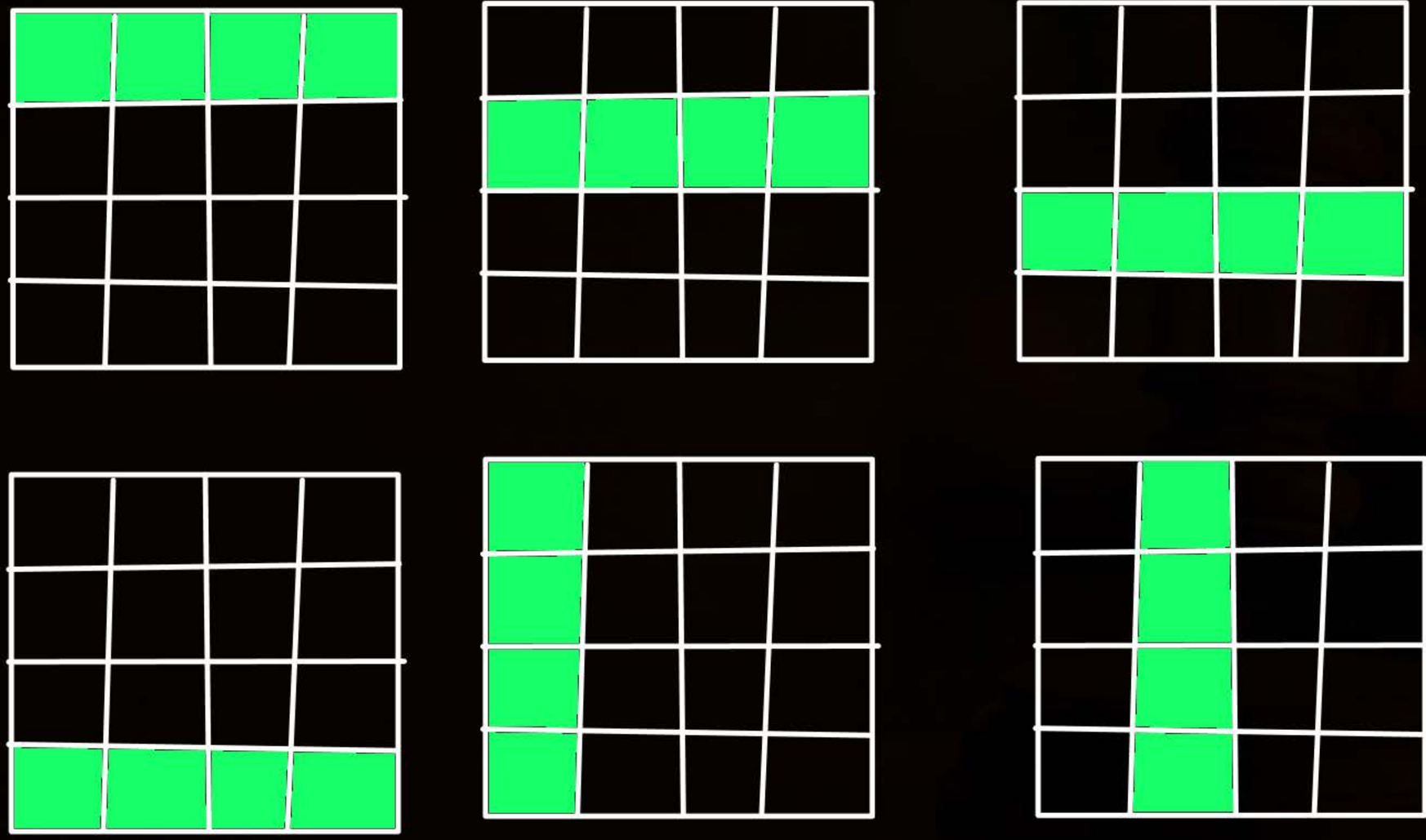




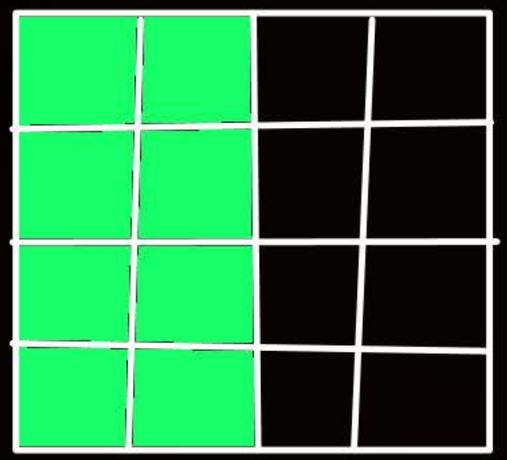


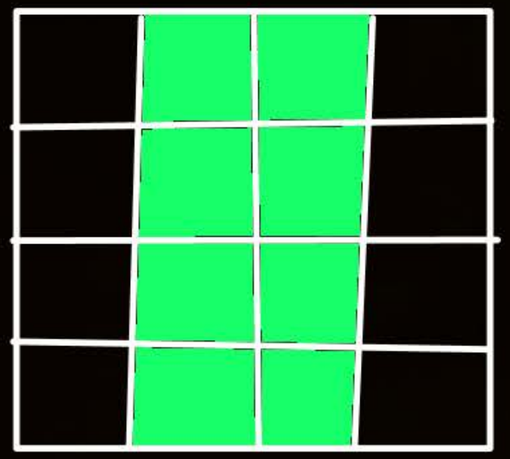


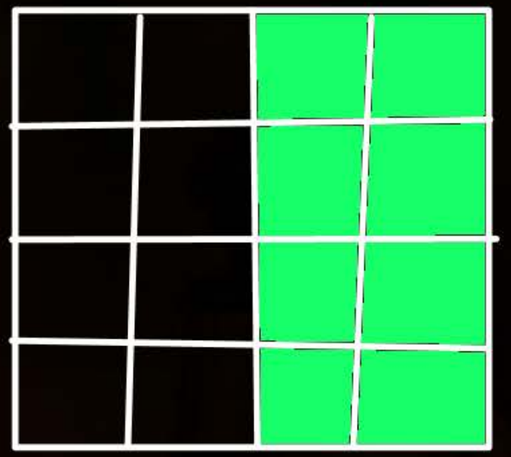







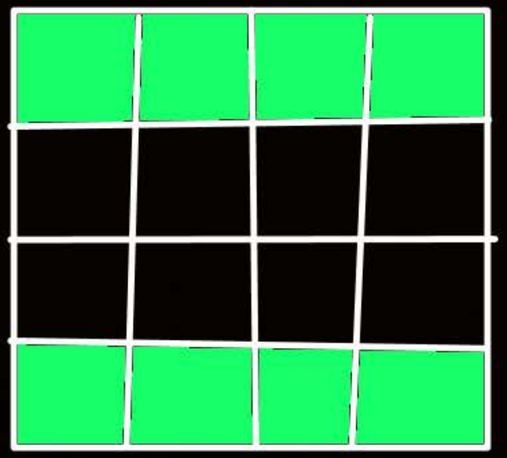


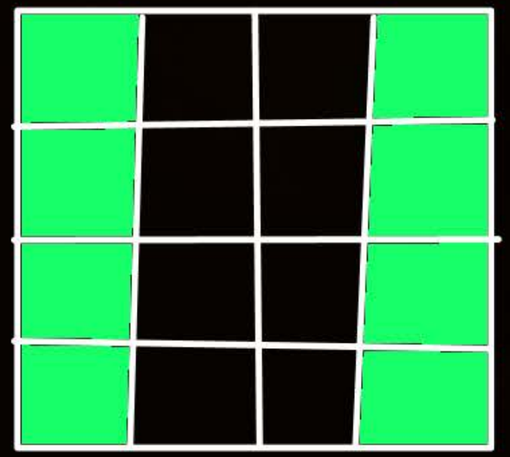
octa

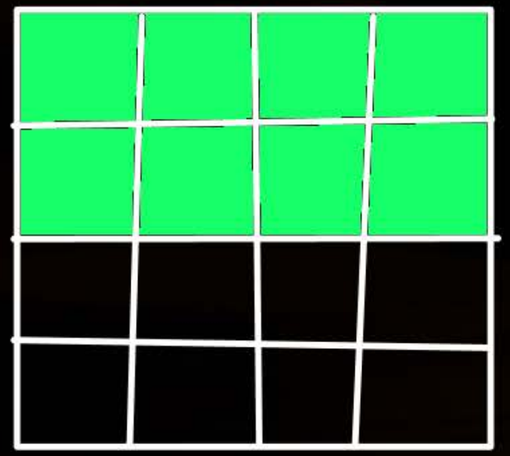




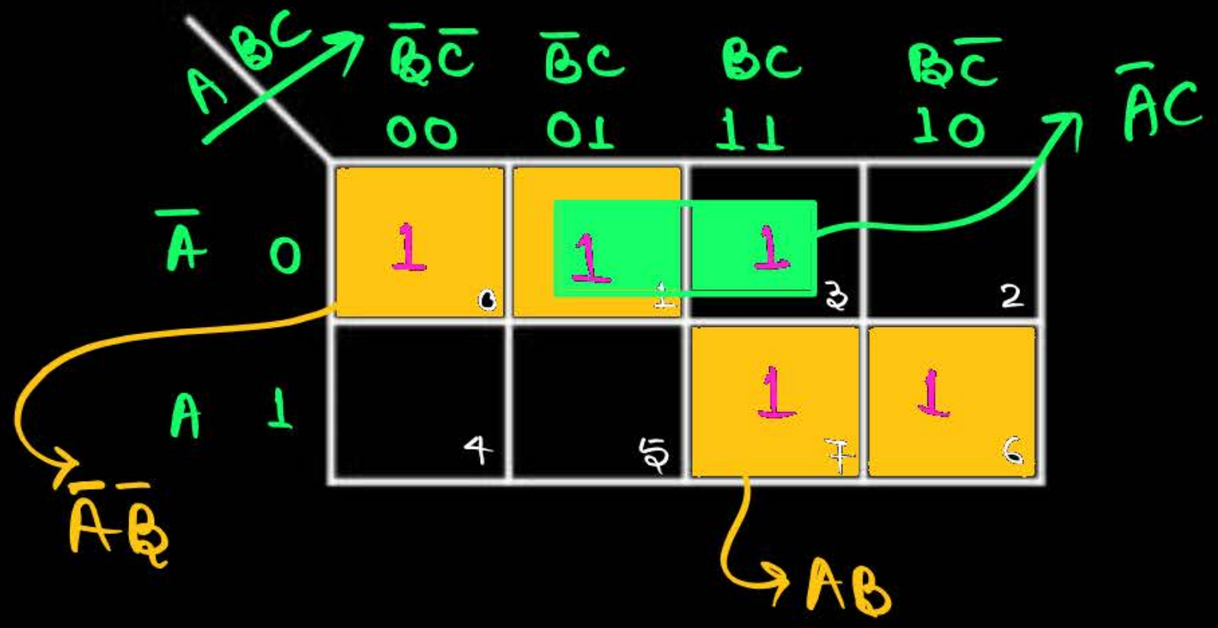




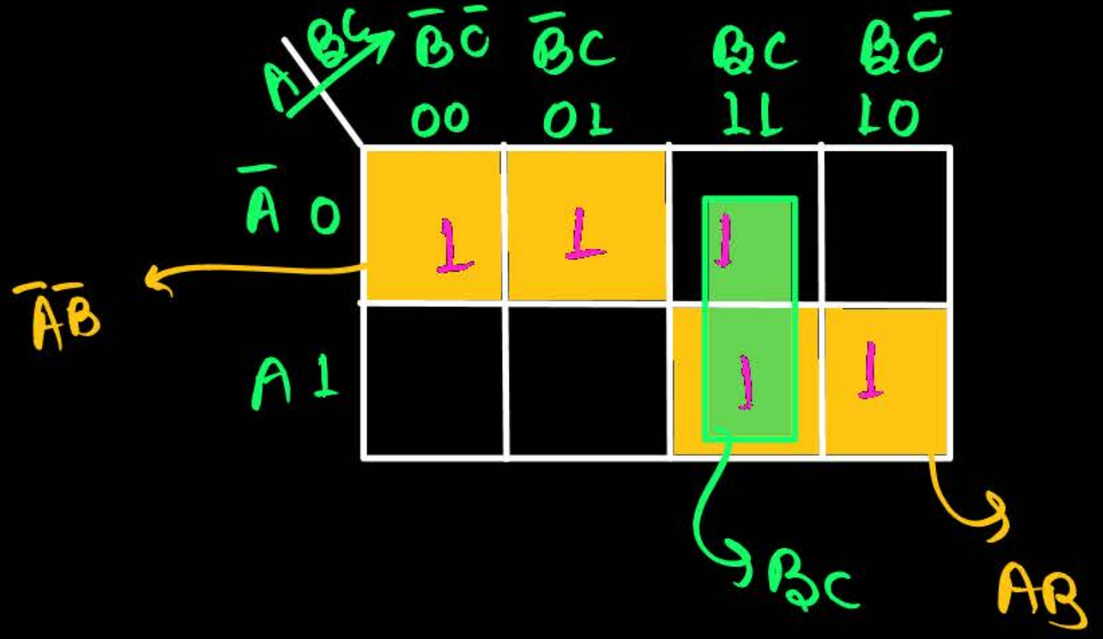




#Q. $f(A, B, C) = \Sigma m(0, 1, 3, 6, 7)$

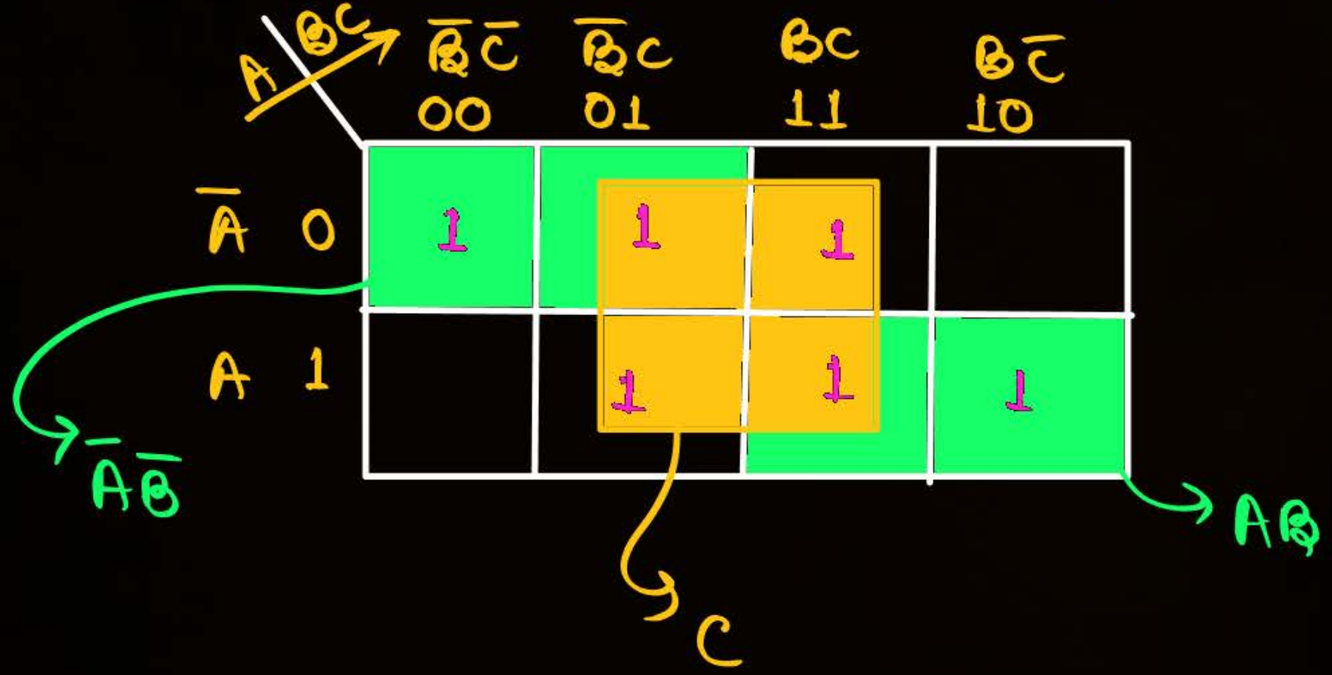


$\bar{A}\bar{B} + AB + \bar{A}C$



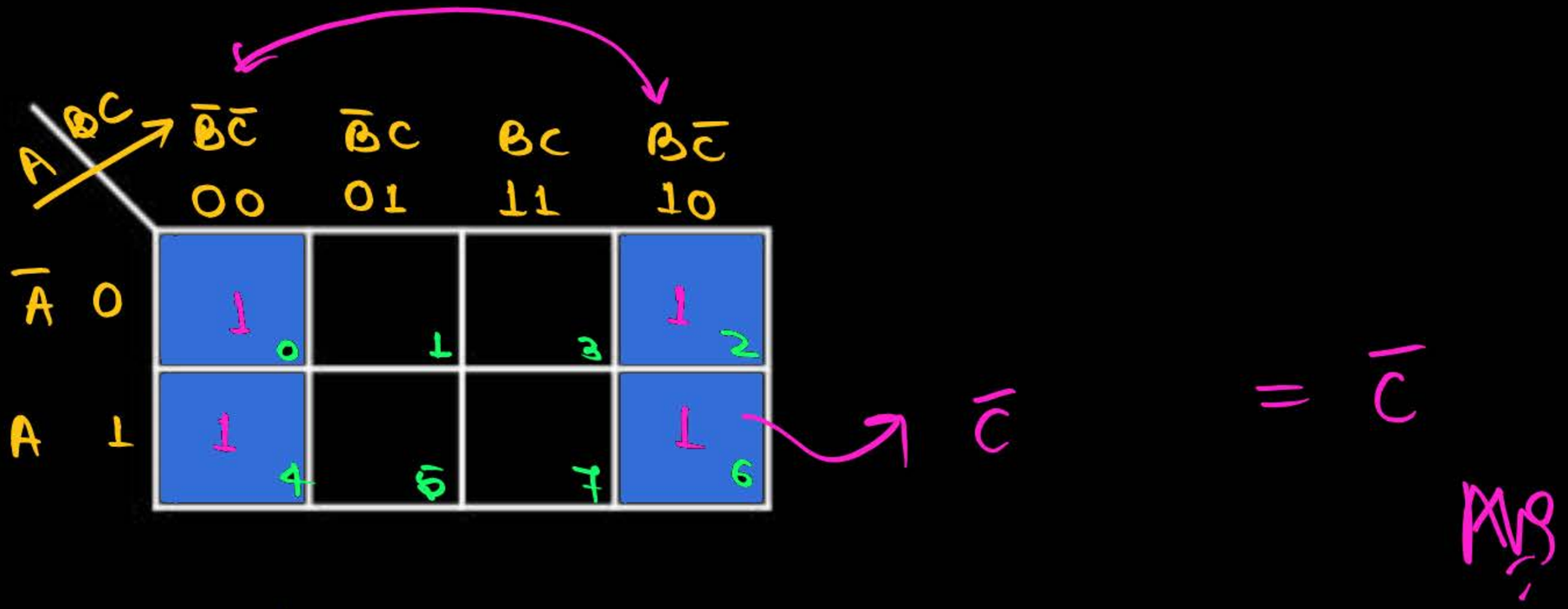
$\bar{A}\bar{B} + AB + BC$

$f(A,B,C) = \Sigma m(0,1,3,5,6,7)$

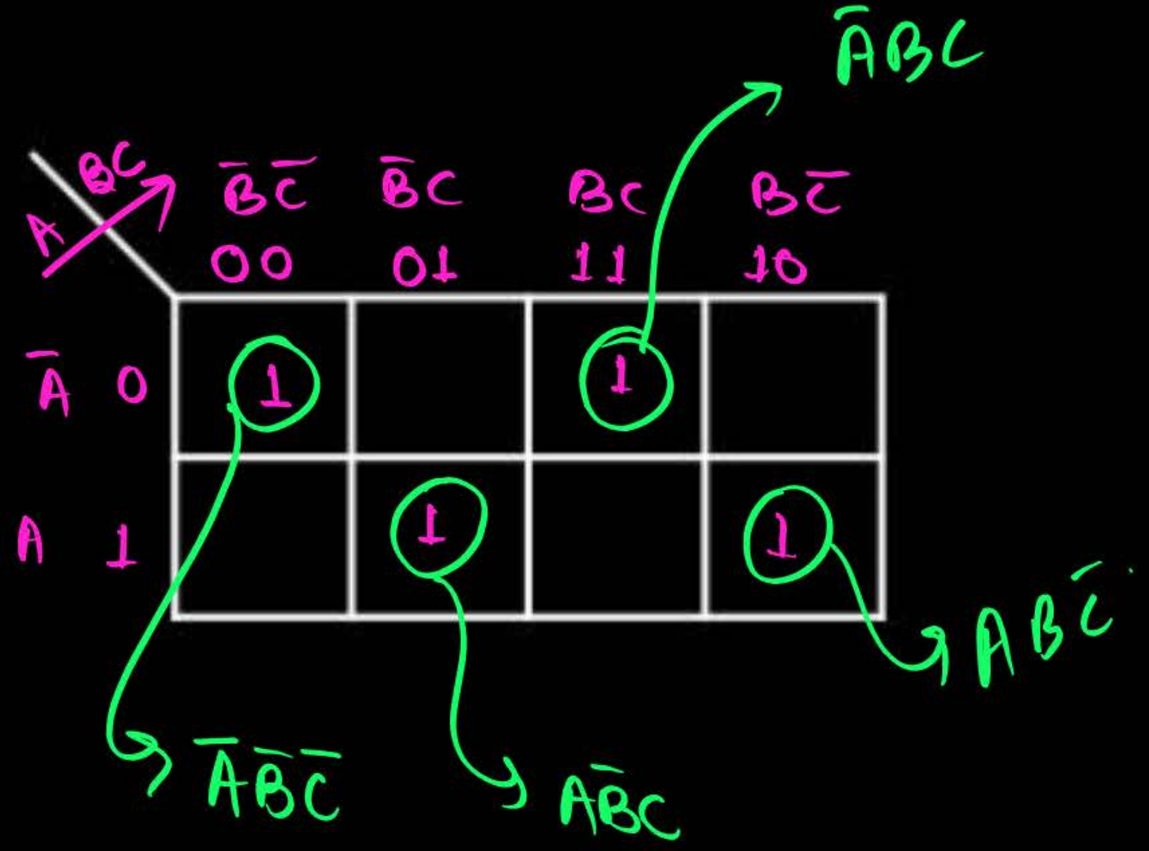


$\bar{A}\bar{B} + AB + C$

#Q. $f(A, B, C) = \sum m\,(0, 2, 4, 6)$



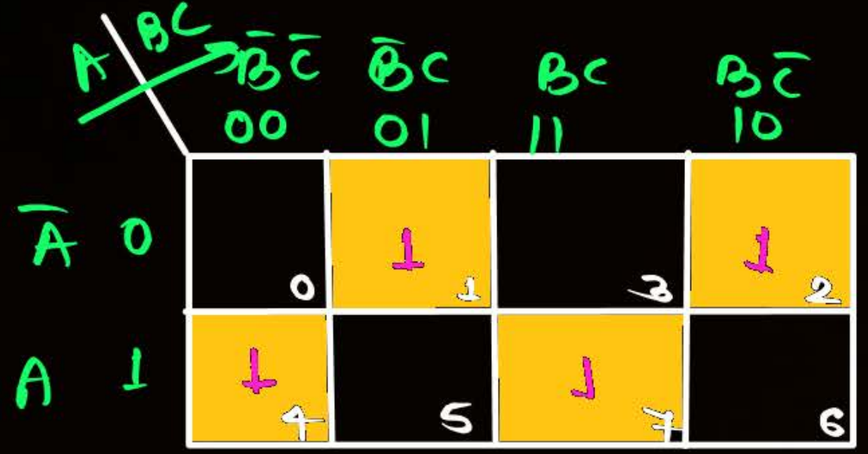
#Q. $f(A, B, C) = \sum m\,(0, 3, 5, 6)$



No minization

already minimized.

$$f(A,B,C) = \Sigma m(1,2,4,7)$$

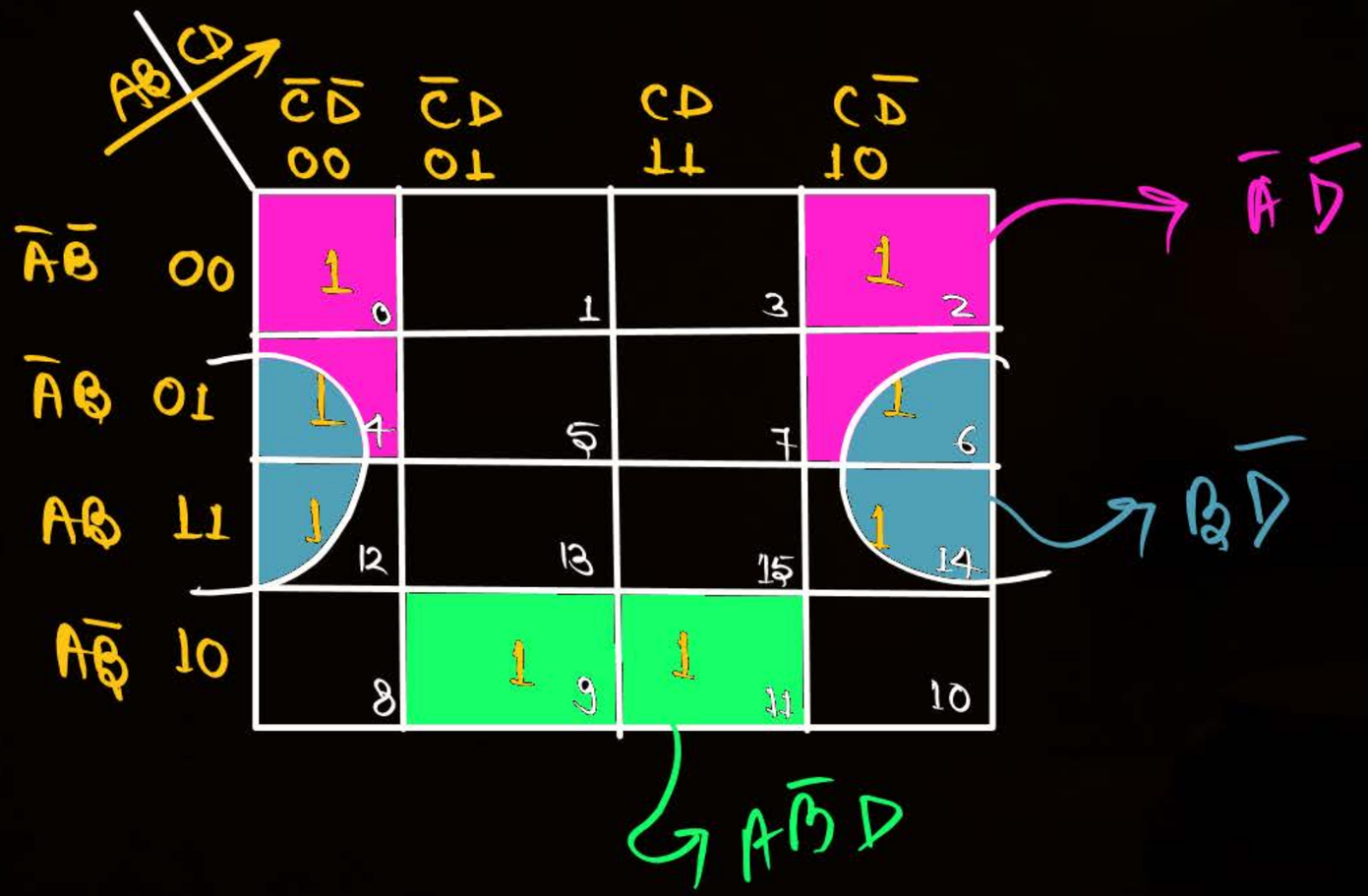


$f(A,B,C,D) = \Sigma m(0,1,4,5,10,14,15)$



$\bar{A}\bar{C} + ABC + AC\bar{D}$

$$f(A,B,C,D) = \Sigma m(0,2,4,6,9,11,12,14)$$



$\bar{A}\bar{D} + B\bar{D} + A\bar{B}D$

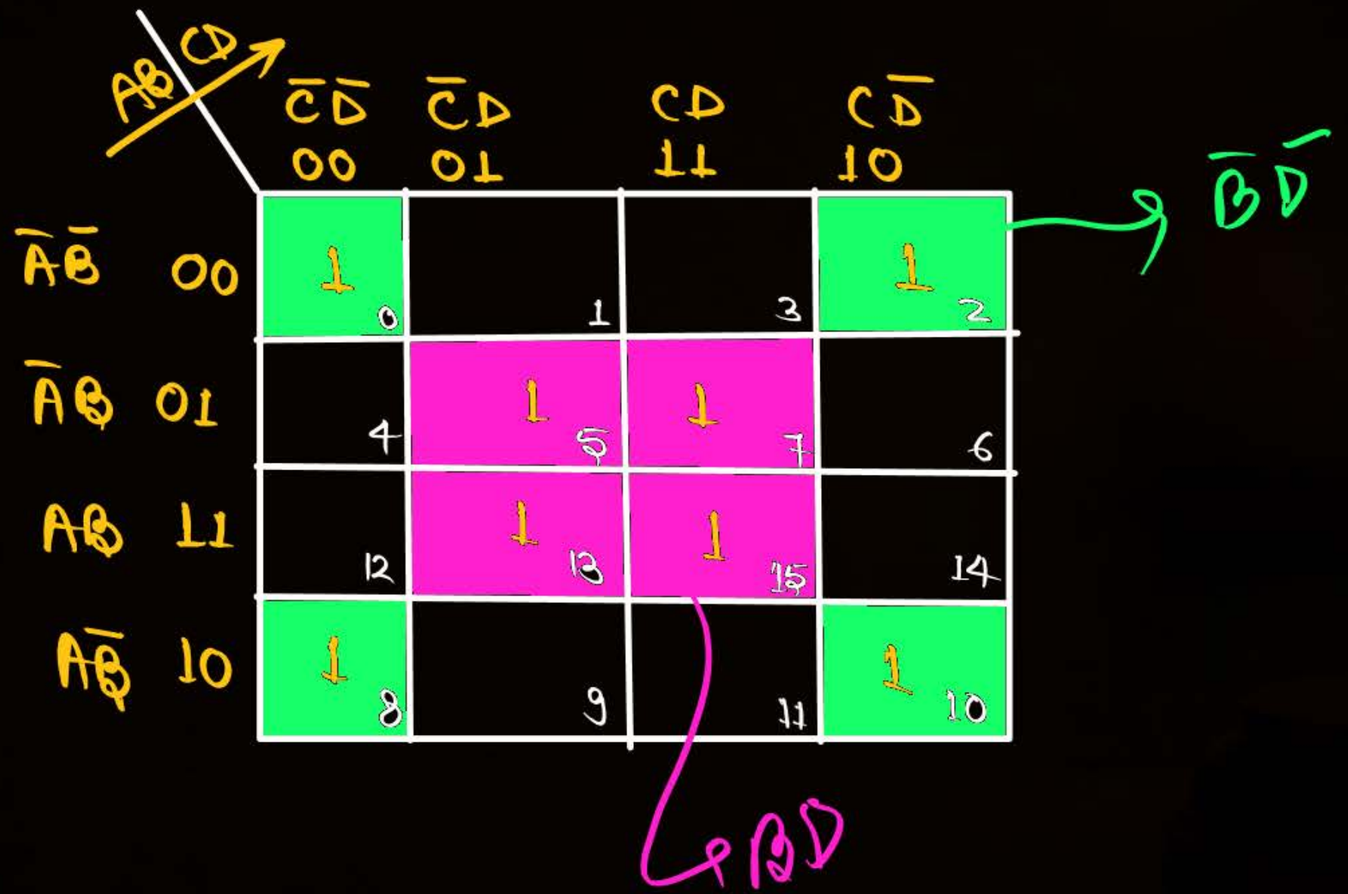
AB

$$f(A,B,C,D) = \Sigma m(0,1,2,3,8,9,10,11,12)$$



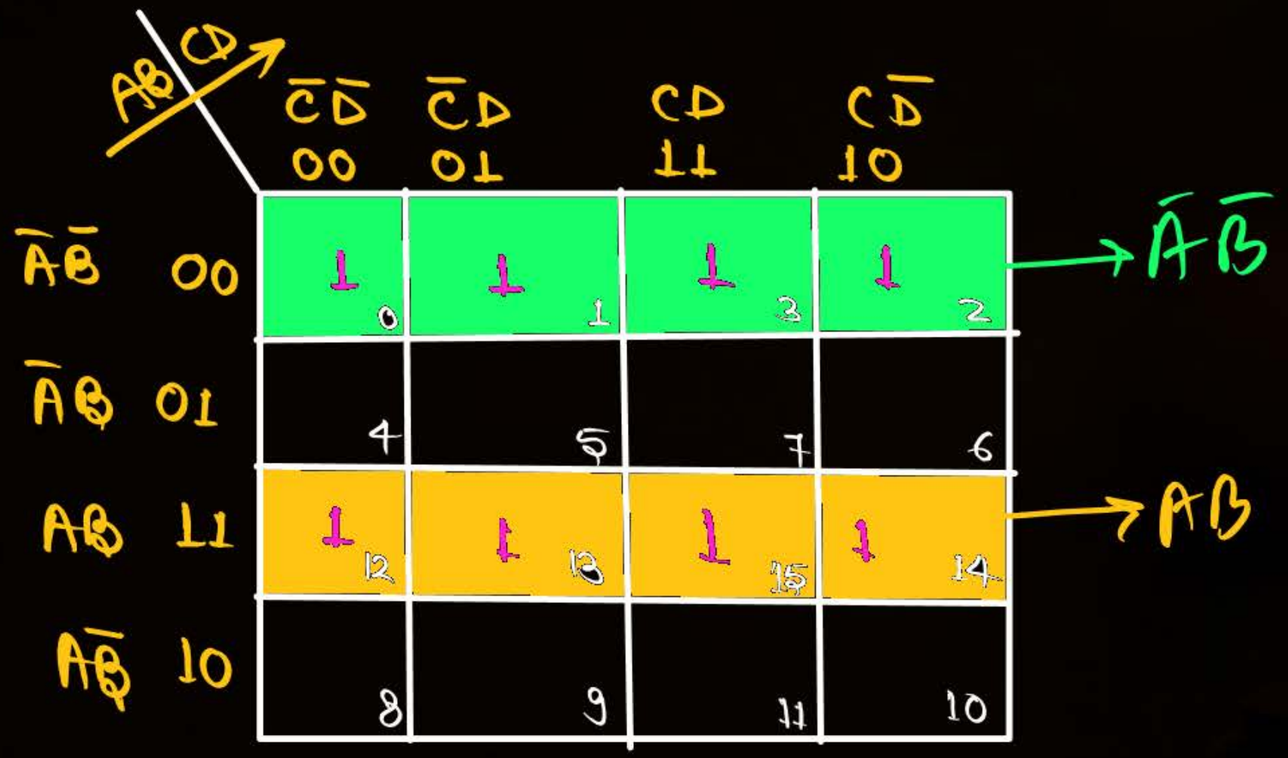
$$\bar{B} + A\bar{C}\bar{D}$$

$f(A,B,C,D) = \Sigma m(0,2,5,7,8,10,13,15)$



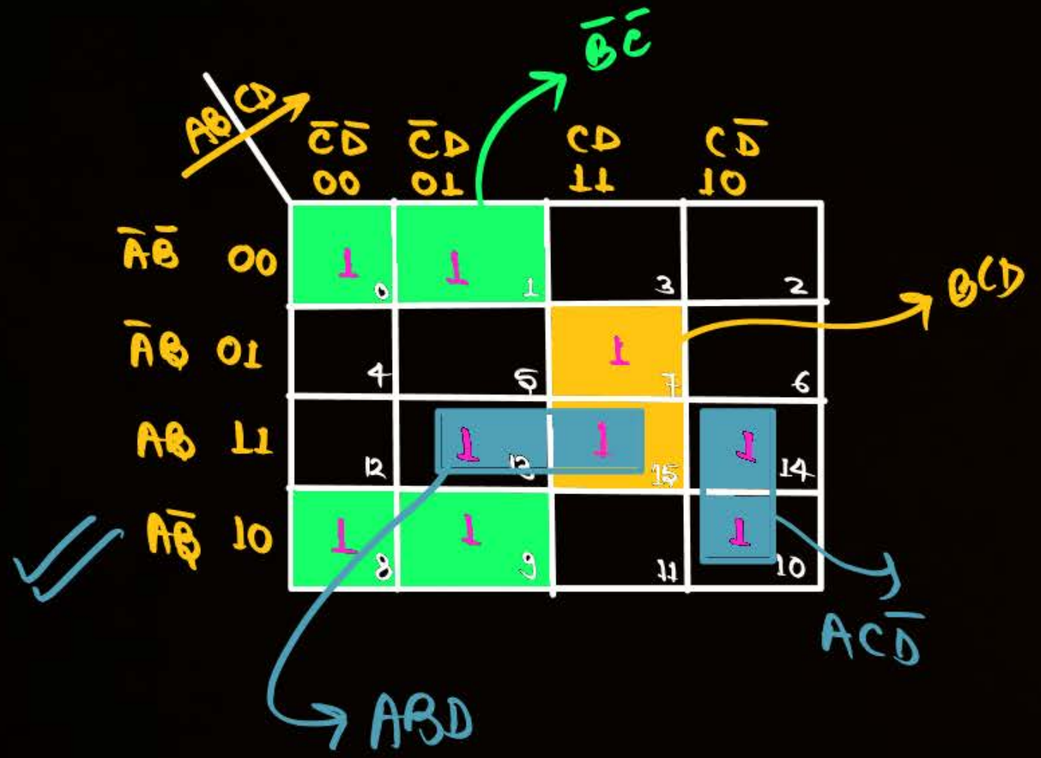
$\bar{B}\bar{D} + BD$

$f(A,B,C,D) = \Sigma m(0,1,2,3,12,13,14,15)$

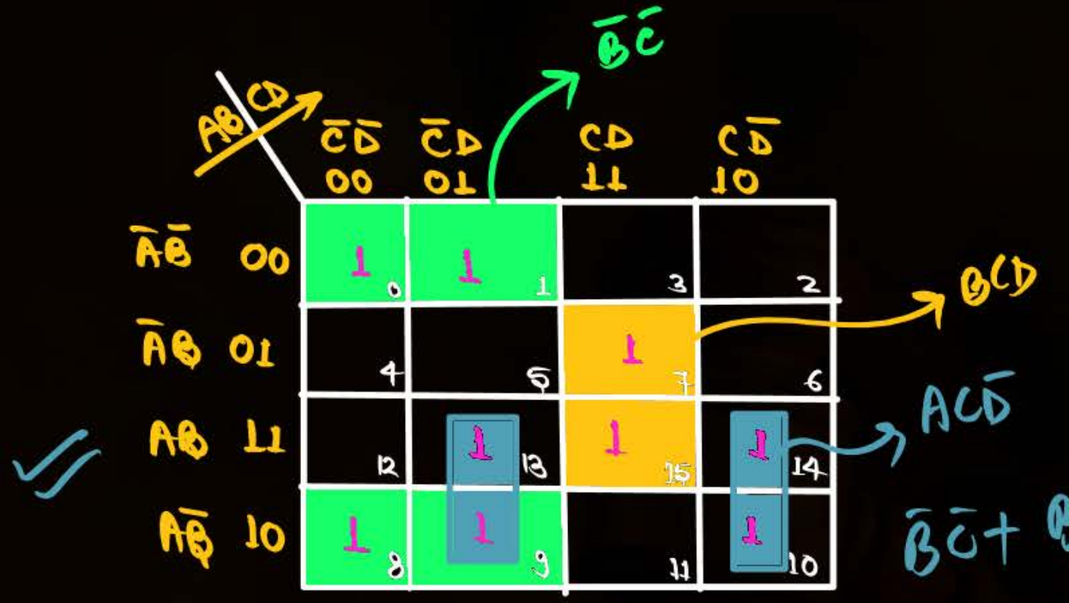


$\bar{A}\bar{B} + AB$

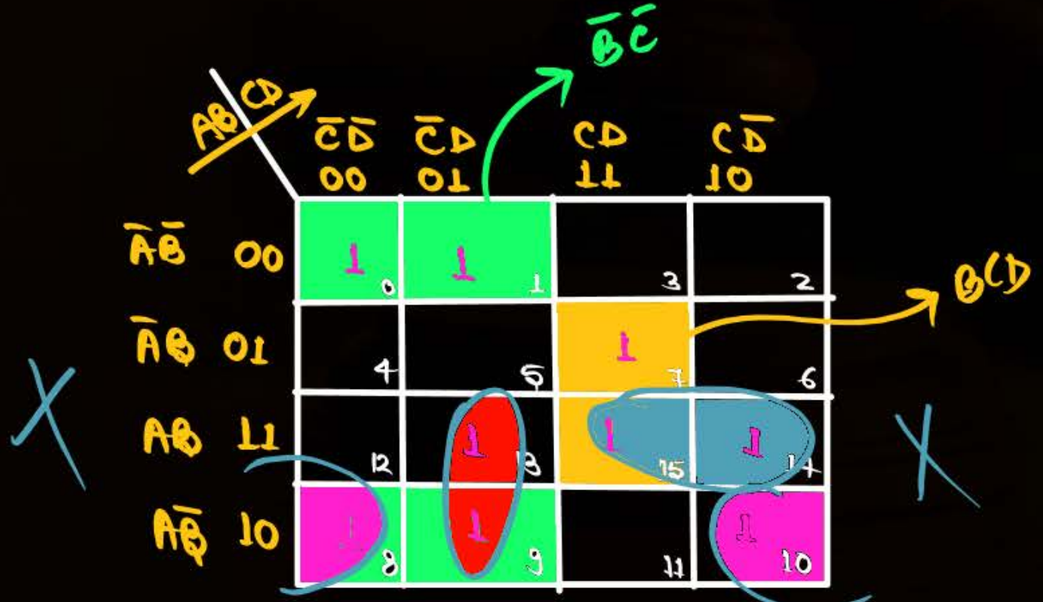
$f(A,B,C,D) = \Sigma m(0,1,7,8,9,10,13,14,15)$



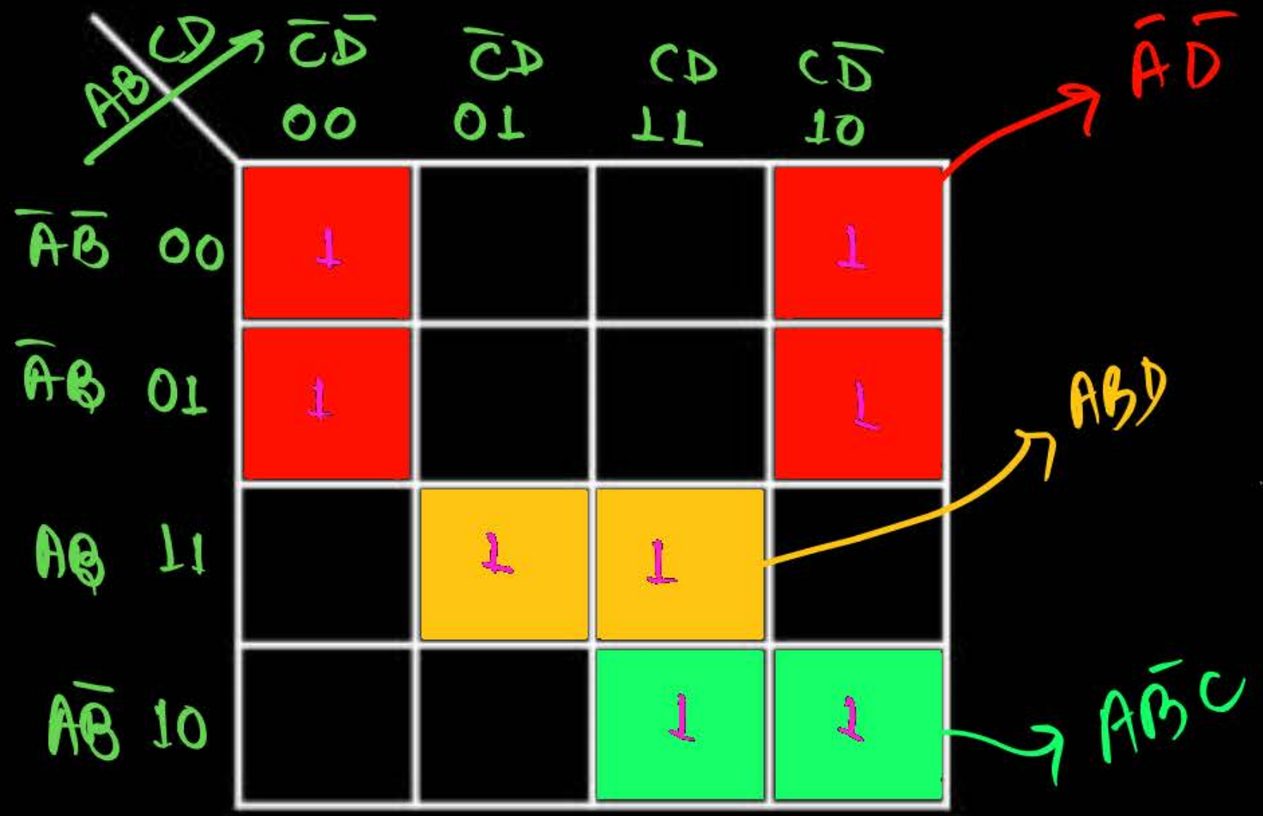
$\bar{B}\bar{C} + BCD + ABD + AC\bar{D}$ ✓



$\bar{B}\bar{C} + BCD + AC\bar{D}$ + $A\bar{C}D$

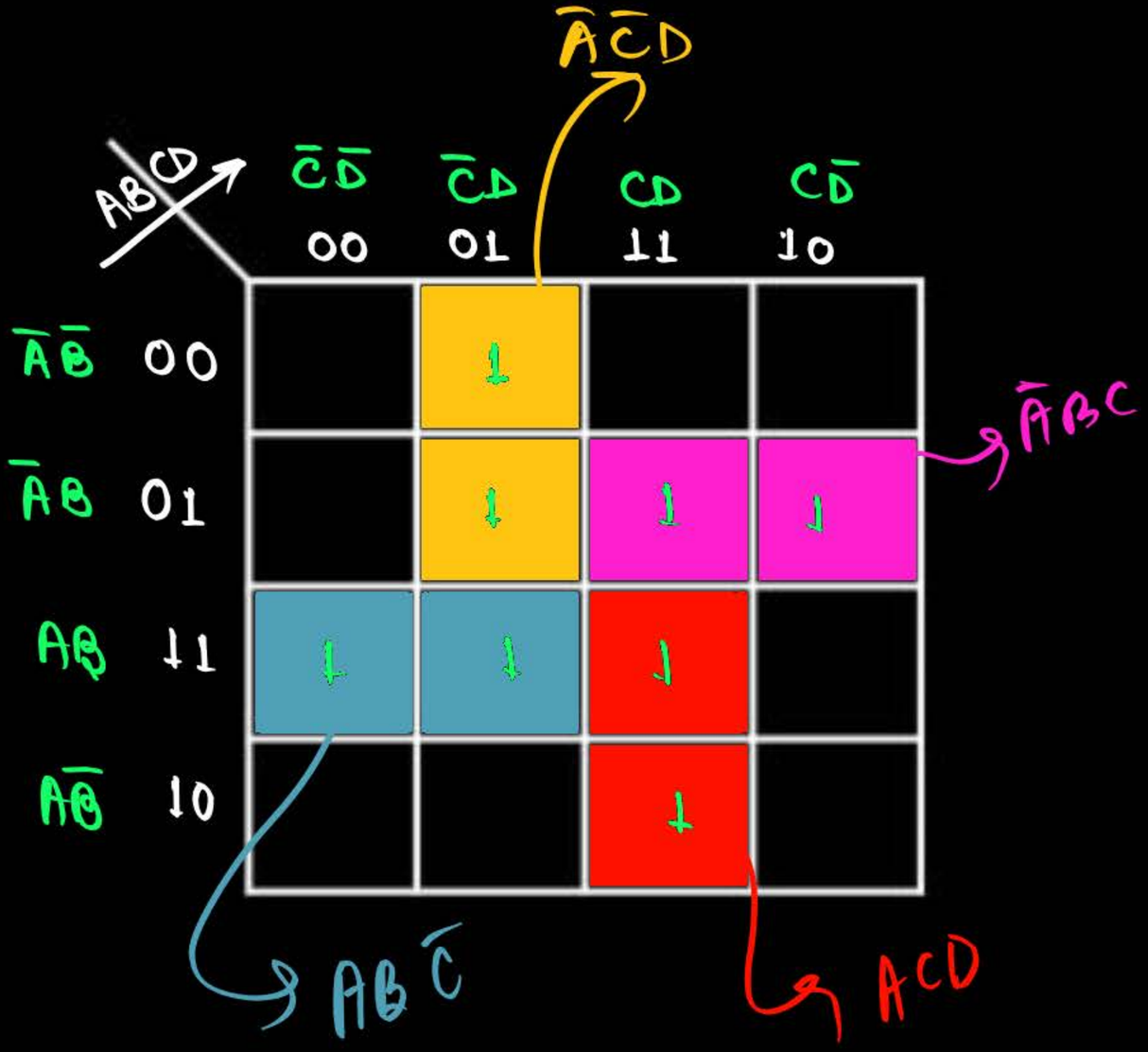


#Q. $f(A, B, C, D) = \sum m\,(0, 2, 4, 6, 10, 11, 13, 15)$



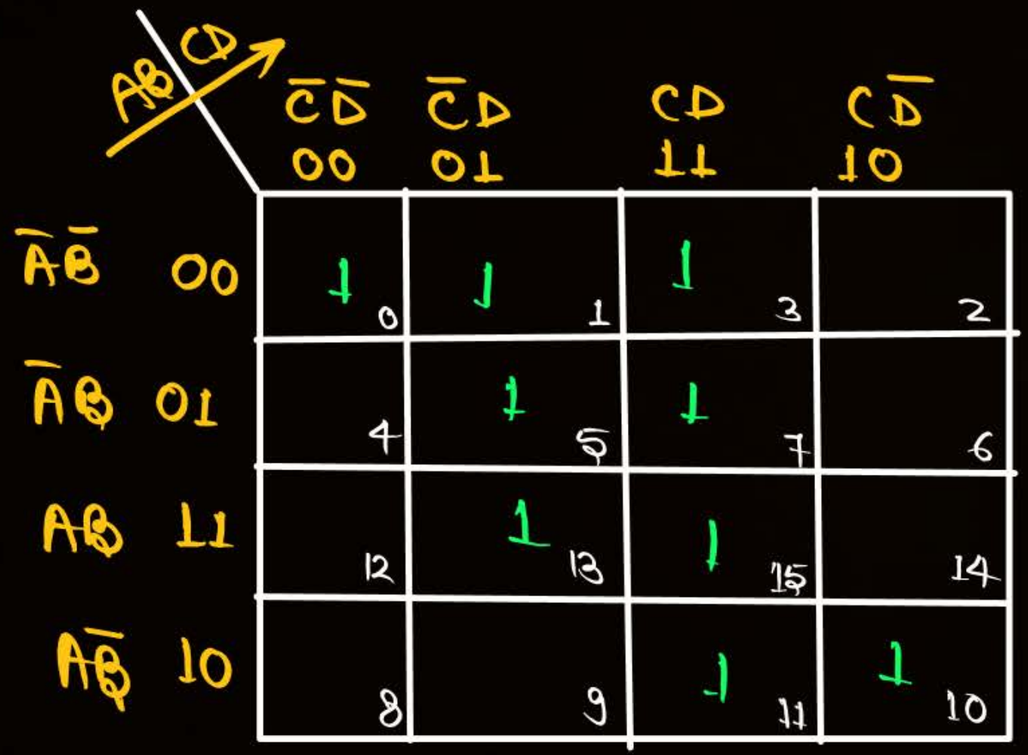
$\bar{A}\bar{D} + ABD + A\bar{B}C$

#Q. $f(A, B, C, D) = \sum m\,(1, 5, 6, 7, 11, 12, 13, 15)$

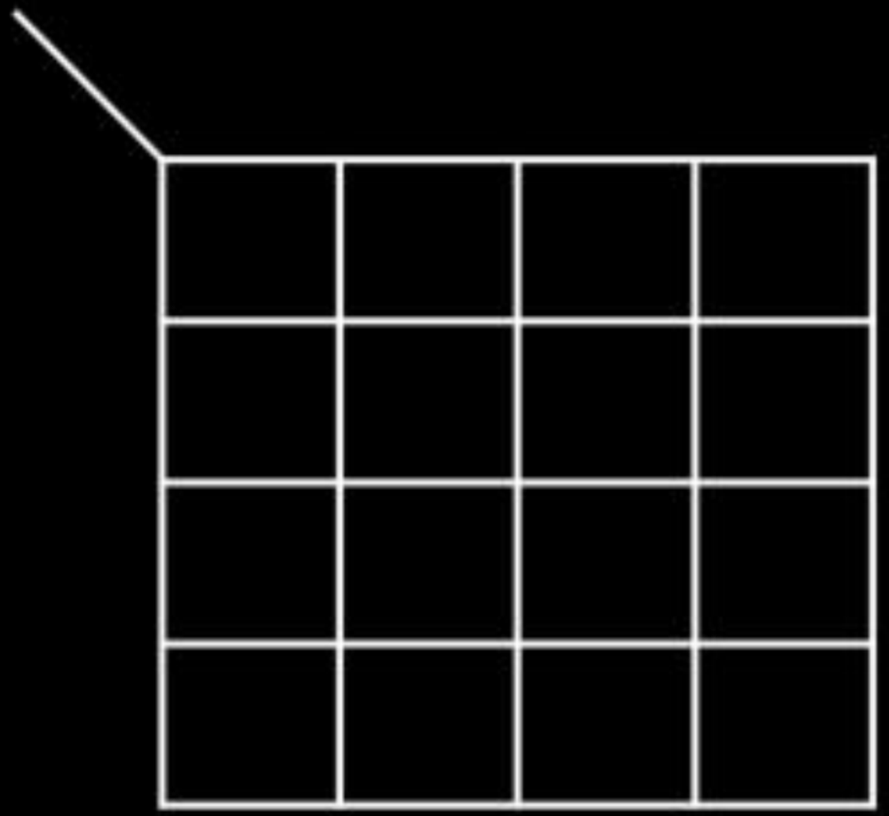


$\bar{A}BC + ACD + AB\bar{C} + \bar{A}\bar{C}D$

$f(A,B,C,D) = \Sigma m($



#Q. $f(A, B, C, D) = \sum m\,(1, 5, 6, 7, 11, 12, 13, 15)$





2 Minute Summary

- Questions Practice
- Discussion

Thank you

GW

Soldiers !

